

ANEMIA

- These are conditions characterized by a **decrease in the number of RBCs or their Hb content or both**, leads to decrease oxygen carrying capacity of the blood. They are classified into:-

A) Hemolytic Anemia:- (due to excessive destruction of RBCs)

- 1- Chemicals (as lead, copper and sulfied poisoning).
- 2- Diseases (as malaria, babesia, snake venoms, bacterial toxins and incompatible blood).
- 3- Increase fragility of RBCs (as in Thalassemia, hemolytic jaundice and sickles anemia).

B) Hemorrhagic Anemia:- (due to excessive loss of RBCs)

1- Acute hemorrhage: as in rapid hemorrhage, hypoxia of the tissues stimulates rapid regeneration of new red cells which are of normal size (normocytic) and contain normal Hb (normochromic). So it called anemia normocytic normochromic anemia.

2- Chronic hemorrhage: as in case of piles, bilharzia, peptic ulcer and uterine hemorrhage.

- The body is deprived of large quantities of iron which decrease Hb synthesis that cause hypoxia which stimulate bone marrow to produce more RBCs but small in size and their Hb content is low. So it named **microcytic hypochromic anemia**.
 - Which is common in **females** (due to loss of iron in cyclic hemorrhage & in milk during lactation and iron transport from the mother to the fetus during pregnancy) and **newborn** (due to the iron is deficient in mother's milk and the iron stores in the young's body is empty within 3-6 months).
 - This anemia can be treated with administration of **iron compound**.
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C) Anemia due to defective RBCs formation:-

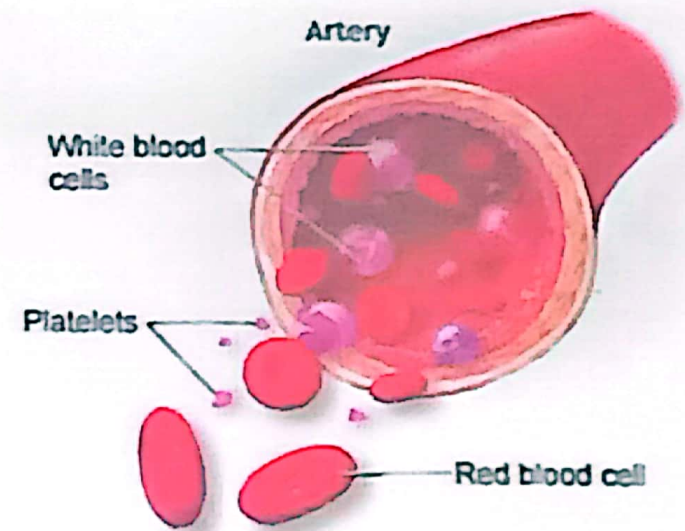
- 1- **Pernicious Anemia**: due to vit.B₁₂ or folic acid deficiency or atrophy of gastric mucosa. This anemia is characterized by:-
 - a- Immature (nucleated), **macrocytic and hyperchromic** RBCs.
 - b- Degenerative **lesions in the spinal cord** and peripheral nerves.
 - c- **Atrophy of gastric mucosa** which causes lack of gastric Hcl, pepsin, intrinsic factor (no absorption of vit.B₁₂) and neuropoietin (a substance important for nerve nutrition, so leads to nerve degeneration).
- Pernicious anemia affected simple stomach animals but rarely affecting ruminant animals (in case of nutritional deficiency of cobalt). This type of anemia can be **treated by vit.B₁₂ injection**.

2- Nutritional Anemia: due to iron, copper or protein deficiency. It is characterized by microcytic hypochromic RBCs and commonly occurs in females and newborn animals.

3- Aplastic Anemia: due to destruction of the bone marrow by prolonged exposure to X-rays or radium, poisoning with industrial chemicals or some bacterial toxins. The red bone marrow is replaced by fatty tissue so, the synthesis of all blood cells is lost gradually and may cause death.

BLOOD PLATELETS (THROMBOCYTES)

- These are **granular** (dense and alpha granules), colorless **non-nucleated** (except birds, fishes & reptiles) **oval bodies** produced by **fetal liver**, **spleen** and **bone marrow**.
- In **adult**, **bone marrow** is the only site of thrombocyte production (from megakaryocyte which not found in birds). **Megakaryocyte** has pseudopodia which detached in blood sinusoid to form blood platelets.



BLOOD PLATELETS (THROMBOCYTES)

- Thrombopoietin secreted from bone marrow, liver, spleen & lung control the formation of thrombocyte and megakaryocyte.
- The diameter of thrombocyte is about $2-5\ \mu\text{m}$ and the life span about 2-5 days. The destructed thrombocytes are removed from the circulation by the reticuloendothelial cells particularly in the spleen.
- The number of thrombocytes is about $250.000-500.000/\text{mm}^3$ in man, $300.000-600.000/\text{mm}^3$ in domestic animals and $25.000 - 40.000/\text{mm}^3$ in birds. It is increase in muscle exercise, day time and hard work animals.

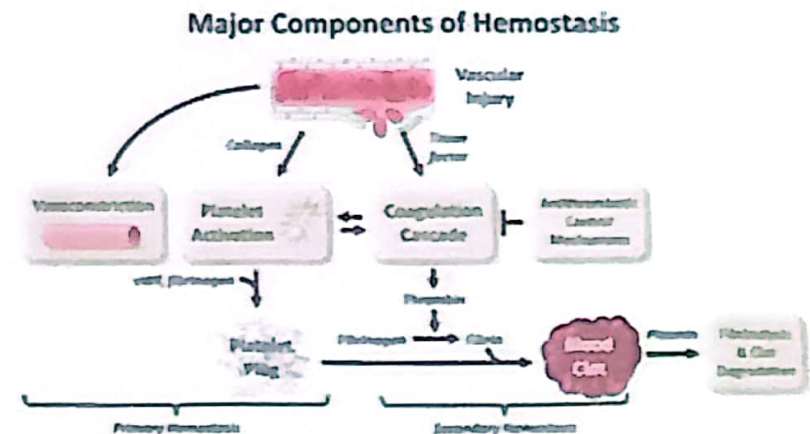
FUNCTIONS OF THROMBOCYTES

A) Hemostasis (stop bleeding):

When the blood vessel wall injured, the platelets interact with exposed collagen and releases the content of their granules which chair in the following steps:-

1- Constriction of the injured blood vessel by a vasoconstrictor substance called serotonin from platelets and also by adrenaline from the sympathetic nerve ending (to decrease the cute opening of the blood vessel).

2- Plug formation: neutrophil secretes thromboxane which aggregate thrombocytes to close the site of injury.



3- **Blood coagulation** (conversion of blood from fluid form to jelly form): the aggregated thrombocytes are disintegrated and releases the clotting factors specially thromboplastin which help blood clotting to seal injured opening of the blood vessel. Also it secretes anti-heparin (to prevent its anticoagulant effect of heparin at the site of injury).

4- **Clot retraction** by thrombocyte retractozyme to adhere the two cutting ends of the blood vessel.

5- **Wound healing** by platelet-derived growth factor (PDGF) which secreted from thrombocyte (also secret from macrophages and endothelial layer of the blood vessel) to stimulate fibroblast multiplication at the site of injury.

- B) Prevent extension of the clot in the neighboring vessel wall:** by prostacyclin (from thrombocyte and intima of the blood vessel) which inhibit platelets aggregation, so prevent thrombus formation.
- C) Phagocytosis:** thrombocytes are engulfing some foreign substances as carbon & viruses.
- D) Defense mechanism:** thrombocytes are secreted antimicrobial substances as β -lysine to hydrolyze the microbes at the site of injury.

Thrombocytopenia:

It is a decrease in the number of thrombocytes leads to deficient in clot retraction and poor constriction of ruptured vessels. This cause a disease called purpura (characterized by multiple subcutaneous pitecheal hemorrhages)

Thrombocytosis:

It is an increase in the number of thrombocytes, which may leads to thrombus formation.

A) CLASSICAL THEORY:-

Fibrin threads forms a network traps blood cells in its meshes, then forms a red jelly-like mass called blood clot. The clot then retracts (due to contraction of fibrin filaments by the **retractozyme**) and squeezes out clear yellow liquid called serum.

↓ **Thromboplastin & Ca^{+2}**

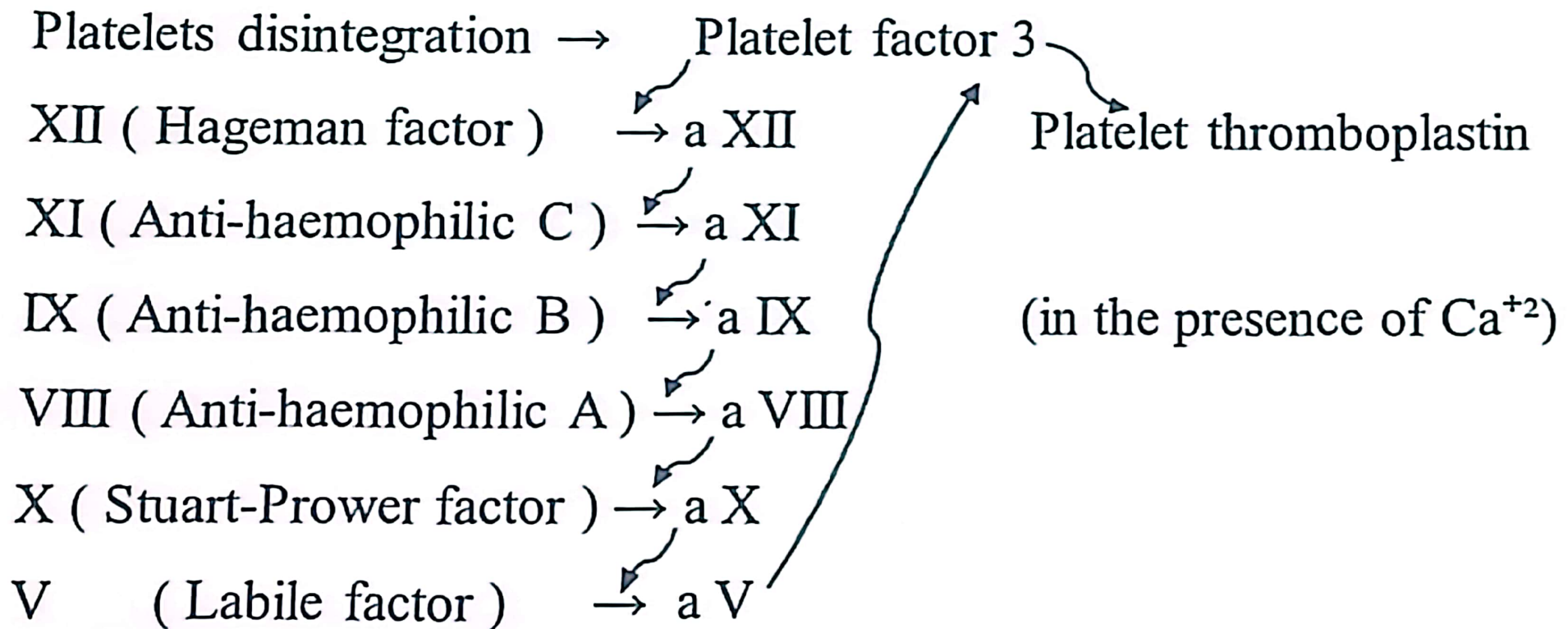
Prothrombin → **Thrombin** ↓

Fibrinogen → **Fibrin**

B) MODERN (ENZYME CASCADE) THEORY:-

1) Formation of active thromboplastins:-

a- Intrinsic (platelet) system of thromboplastin: (4-8min.)



B) MODERN (ENZYME CASCADE) THEORY:-

1) Formation of active thromboplastins:-

b- Extrinsic (tissue) system of thromboplastins: (12-20 seconds)

Tissue damage $\rightarrow$ Tissue factor $\rightarrow$ Tissue thromboplastin
VII (Stable factor) $\xrightarrow{\text{tissue factor}}$ a VII (in the presence of Ca^{+2})
X $\xrightarrow{\text{a VII}}$ a X

The three pathways that make up the classical blood coagulation pathway

Intrinsic

surface contact

XII $\xrightarrow{\quad}$ XII_a

XI $\xrightarrow{\quad}$ XI_a

IX $\xrightarrow{\quad}$ IX_a

(VIII, PL, ↓ Ca⁺⁺)

X $\xrightarrow{\quad}$ X_a

(V, PL, ↓ Ca⁺⁺)

prothrombin $\xrightarrow{\quad}$ thrombin (serine protease)

fibrinogen $\xrightarrow{\quad}$ fibrin

XIII_a

stable fibrin clot

- XII – Hageman factor, a serine protease
 - XI – Plasma thromboplastin, antecedent serine protease
 - IX – Christmas factor, serine protease
 - VII – Stable factor, serine protease
 - XIII – Fibrin stabilising factor, a transglutaminase
 - PL – Platelet membrane phospholipid
 - Ca⁺⁺ – Calcium ions
 - TF – Tissue Factor
- (_a = active form)

Extrinsic

TF·VII_a ← tissue damage

X $\xrightarrow{\quad}$ X_a

Common

XIII

↓

XIII_a

B) MODERN (ENZYMIC CASCADE) THEORY:-

- # All the clotting factors are found in inactive form until needed.
- # There is no factor VI.
- # Coagulation becomes rapid once factor V is activated.
- # There is a species difference in the level of clotting factors.
- # Factor XII is absent in marine mammals, reptiles and birds (in which XI is absent also).
- # Clotting time progressively longer in the following order: cat, dog and pig, then horse and sheep then cow and finally birds.

IN HEALTHY CONDITIONS, BLOOD DOES NOT CLOT INSIDE THE BLOOD VESSELS BECAUSE:-

1) Heparin: It is an anticoagulant (used in vivo and vitro) secreted from mast cells and basophils.

- Mechanism of action: Stabilizing the thrombocytes, antithromboplastin, antiprothrombin, antithrombin and fibrinolysis.

2) Plasmin (fibrinolysin): It is a proteolytic enzyme secreted from eosinophils in inactive form plasminogen. It is increased after emotional stress, surgical operations and epinephrine secretion. It activated to plasmin by protein C activator, factor XII and urokinase enzyme. Liver produces antiplasmin to inhibit its action.

- Mechanism of action: Antithrombin, breakdown of factor I, V & VIII and fibrinolysis.

- Important of fibrinolysis:-

- a- Keep fluidity of blood inside the blood vessels by removing clots occluding them.
- b- Keep fluidity of menses blood, so preventing infection and remain the female genital passage open.
- c- It helps rapid removal of any blood clots formed in the tissues.

3) The clotting factors found in the plasma in inactive form.

4) Removal of activated clotting factors by the liver after finishing its action.

5) Endothelial lining of blood vessel have many effects:-

a- It has a very smooth surface, which sliding the thrombocyte and preventing its aggregation and its disintegration & so prevent clot formation.

b- It has -ve charge as that of thrombocyte, so repelling it and preventing its aggregation.

c- It secrete thrombomodulin, which convert thrombin to protein C activator. So, prevent formation of fibrin, inactivate factor V & VIII and activate plasminogen. Thrombomodulin increased in diabetic individual.

d- It secrete prostacyclin, which prevent platelets aggregation.

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- 6) Flow of blood in the blood vessel is not slow.
 - 7) Optimal number of platelets & Clotting factors levels.
 - 8) Presence of anti-thrombin III in the plasma, which prevent thrombin formation.
 - 9) Prostacyclin secreted from platelets and intima of the blood vessel, which prevent platelets aggregation. Aspirin, tranquilizer and local anesthesia have the same effect on the platelets.

DISORDERS OF COAGULATION

1) **Vit. K deficiency:** vit. K is important for the formation of factor II (prothrombin), VII, IX and X by the liver, so its deficiency inhibits their formation which causes a tendency for hemorrhage.

- **Causes of vit.K deficiency:-**

- 1- Its deficiency in the food.

- 2- **Obstructive jaundice** (i.e. no bile, so leads to lack of vit.K absorption which is a fat soluble vit.).

- 3- **Dicoumarol** (as in spoiled hay) causes competitive inhibition of vit.K. Dicoumarol is used in cases which have a tendency to intravascular clot as in old age and long recumbency. Also it is used as a rodenticide.

DISORDERS OF COAGULATION

2) Hemophilia: Inherited disease transmitted by females (which not affected but act as a carriers) to males. A slight injury causes sever hemorrhage which may continue for several hours. Coagulation time is 1-12 hours.

- **There are 3 types of hemophilia:-**

- 1- **Hemophilia A:** due to lack of factor VIII. The more common type in man and animals, it causes a hemorrhage in the muscles which leads to fibrosis in the muscle tissues and cause semi-paralysis.

- 2- **Hemophilia B:** due to lack of factor IX, not common type.

- 3- **Hemophilia C:** due to lack of factor XI, it found in man, cattle and dog.

- **Treatment:** by injecting the deficient factor.

DISORDERS OF COAGULATION

3) **Thrombosis (intravascular clotting)**: Any disease cause roughness in the inner surface of the blood vessel or long recumbency cause stick of platelets which leads to its disintegration and clot formation. Intravascular clot flow with blood in the form of clot emboli till closed a fine blood vessel to cause clot thrombi.

- **Side effects of thrombosis:-**

- a- If occurs in veins, increase the venous pressure and filtration in this place and cause local edema.

- b- If occurs in arteries, leads to ischemia of the tissues supplied by it and stoppage of its functions. If this occurs in the brain centers may cause paralyses, blindness, deafness,.....

DISORDERS OF COAGULATION

Prevention of the thrombosis:-

- a- Prevent prolonged recumbency (or use air bed or body massage to increase the blood flow).
- b- Use of anticoagulant (as dicoumarol) in cases has a tendency to thrombosis.

DISORDERS OF COAGULATION

4) Purpura: It is a disease caused in thrombocytopenia or due to increase the fragility of the blood vessels, leads to small hemorrhage (petechial) in the skin and mucous membranes. Bleeding time is long (15-20 minutes), clot is soft and poorly retracted.

5) Von-Willebrand disease: It is an inherited bleeding disease (like hemophilia A) affects both male and female man and pig (affected pigs may bleed till death), it causes uterine hemorrhage in woman.

	Heparin	Dicoumarol
1- Origin	Animal	Plant
2- Work	Rapidly(after few minutes)	Slowly (after 1-2 days)
3- Duration of work	Hours	Days
4- Mechanism of action	Anti-prothrombin ; anti-thrombin... & fibrinolytic	Inhibit production of factors 2; 7; 9 & 10 by the liver (by competitive inhibit vit.k
5- Site of action	In vivo & in vitro	In vivo only
6- Route	I/V or I/M	Orally
7- Anti-dote	Protamine sulphate 1%	Vit.k